SE(3)-WAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation

Anonymous CVPR submission

Figure placeholder: Fig. 1 teaser — a role-aware 4D view of a moving scene (real robot)

Layout: full width, two parts. **(a) Left, ∼62%:** one real-robot frame of the *linear-motion grasp* at the instant a chunk is issued. Overlays in the paper's semantic colors: blue = the target's past SE(3) poses from the causal tracker (small triads at $-350 \ldots 0$ ms, fading with age); orange = the target's SE(3) forecast at the execution times of the chunk, taken from the training-time heads evaluated offline; light orange arrows = sparse 3D flow on the context on the 14×14 grid only; purple = the commanded gripper path meeting the target at τ^{a}. A gray ghost marks the pose observed at capture, with a dashed red arrow showing where a reactive policy would close. A thin timeline strip under the image: capture $\tau_t \rightarrow$ latency $\ell_t \rightarrow$ execution τ^{a}. **(b) Right, ∼38%:** top: "Image-space WAM: future frames or latents at uniform fidelity" (filmstrip icon); middle: "SE(3)-WAM: SE(3) for the target, sparse 3D flow for the context, training only" (triads + sparse arrows); bottom **(required)**: SR vs. added latency (0–200 ms) for SE(3)-WAM and the strongest baseline (data: the latency sweep of Appendix E.2). **Rules:** real photo kept unretouched; overlays drawn from logged tracker/forecast/command data of that trial; caption states the trial's outcome.

Figure 1. **A role-aware 4D view of a moving scene.** Between image capture and the first command, a part on the conveyor moves farther than the grasp tolerates. SE(3)-WAM encodes the target's metric motion (blue) and is trained to forecast the target's SE(3) pose precisely and the surrounding scene as a sparse 3D flow (orange), a forecast it skips at deployment, so that its commands meet the target when they execute (purple). Image-space world–action models forecast appearance at uniform fidelity instead. Right: success against added latency on Dynamic-14.

Abstract

Manipulating moving objects requires actions that fit the scene at the moment they execute, not when the image was captured. A policy must therefore infer how the scene moves and anticipate where it will be, yet every millisecond spent on this inference lets the target move further. Existing methods resolve this tension poorly: world–action models forecast future images or latents at uniform fidelity, spending capacity and time on appearance while leaving the target's metric pose implicit, and target-centric predictors ignore the surroundings that constrain the motion. We observe that the parts of a dynamic scene differ in the precision they require: the target's pose determines contact to within millimeters, whereas the rest of the scene matters only through the clearance it leaves. We present SE(3)-WAM, a world–action model that learns this role-aware 4D structure. It encodes the target's motion as a metric SE(3) pose history and is trained to forecast the target's future SE(3) trajectory precisely and the surrounding scene coarsely as a sparse 3D flow. Its action stream reads the representation shaped by these forecasts and generates action chunks by conditional flow matching, replanning after every step; the forecasts themselves are never fed back to the actions and are skipped at deployment, so inference needs neither past frames nor a generated future and stays fast enough for real-time control. On Dynamic-14, a simulated benchmark in which the scene keeps moving during inference, SE(3)-WAM achieves [XX.X]% success, [XX.X] points above the strongest baseline, and it succeeds in [XX]% of trials across five real-robot tasks with linear, circular, and accelerating target motion.

1. Introduction

Learning-based manipulation has advanced rapidly, from diffusion policies to vision–language–action (VLA) and world–action models (WAMs) [9, 16, 36, 54, 72, 81, 103], yet most of this progress assumes that objects stay still unless the robot moves them. In factories, kitchens, and human–robot handovers, objects keep moving while the robot perceives, computes, and acts. This poses two coupled difficulties. A single image does not reveal how the target is moving, and by the time an action executes, the target is no longer where it was seen: a part on a conveyor at 0.1 m/s travels 15 mm during a 150 ms delay, more than a parallel-jaw grasp typically tolerates (Fig. 1).

Recent work equips policies with a memory of past motion and a forecast of the future [19, 48, 65, 91]. Both are costly in the resource that dynamic manipulation lacks most: time. Richer histories and denser forecasts add computation, and computation adds latency, which enlarges the very error the forecast is meant to remove. Existing designs therefore trade information against speed. Video-based WAMs sit at

one end [48, 72, 81]: they forecast whole future frames or latents at uniform fidelity, pay for it in latency, and spend most of their capacity on appearance, while the quantity that decides contact, the target's metric pose when the action executes, remains implicit. Methods that forecast only dynamic regions or salient tokens [65, 91] reduce where they predict but still predict in image space, where that pose remains implicit as well. Target-centric predictors sit at the other end [19, 27, 49]: they are fast but blind to surroundings that can block or redirect the motion. What none of them provides is knowledge of the moving scene that is precise where contact depends on it, coarse elsewhere, and cheap to use at execution time.

Our key observation is that a dynamic policy needs to know very little precisely. Contact depends on the target's pose at the moment the action executes, to within millimeters, whereas the rest of the scene matters only through the clearance it leaves. Nor does the forecast have to be computed at test time: a forecast that supervises training shapes what the policy represents and can be dropped at deployment. Following this observation, we present SE(3)-WAM, a role-aware world–action model for dynamic manipulation (Fig. 2). Its input keeps only what motion requires: the current image, encoded by a frozen encoder, and short histories of the target's metric SE(3) pose and of the robot state, which convey motion directly instead of a stack of past frames. A world stream condenses these inputs into a small set of summary tokens and is trained to forecast, in one metric 4D space, the target's future SE(3) trajectory precisely and the rest of the scene as a sparse 3D flow. An action stream reads the summary tokens at every layer through shared attention, generates action chunks by conditional flow matching, and replans after every executed step, so its actions follow the target as it moves. The forecasts act only as supervision: they shape the summary tokens during training, are never fed back to the actions, and their heads are skipped at deployment. SE(3)-WAM thus gains the precision of a metric forecast at a latency close to that of a reactive policy.

We evaluate SE(3)-WAM on Dynamic-14, a simulated benchmark of fourteen tasks in which observation, scene motion, and action execution share one physical clock, so the scene keeps moving while a policy computes. SE(3)-WAM achieves state-of-the-art success of [XX.X]%, [XX.X] points above the strongest baseline, at a p95 latency of [XX] ms, and succeeds in [XX]% of trials across five real-robot tasks with linear, circular, and accelerating target motion, against [XX]% for the strongest baseline. Ablations show that the SE(3) pose history outperforms visual histories, that role-aware forecasts outperform target-centric and image-space forecasts, and that using the forecast only as a training signal matches feeding it to the actions while keeping deployment fast.

Our contributions are threefold:

- We identify that in dynamic manipulation, inference latency turns directly into contact error, and propose role-aware 4D modeling, which represents the target precisely by its metric SE(3) pose history and future trajectory and the rest of the scene coarsely by a sparse 3D flow.
- We present SE(3)-WAM, a world–action model whose action stream reads a representation shaped by these role-aware forecasts, while the forecasts themselves are never fed back and are skipped at deployment, obtaining the precision of prediction at a latency close to that of a reactive policy.
- We build Dynamic-14, a benchmark in which perception, scene motion, and execution share one physical clock. SE(3)-WAM achieves state-of-the-art results on it and on five real-robot tasks, and ablations confirm the contribution of each component.

2. Related Work

Dynamic manipulation. Acting on a moving target requires compensating for the time between sensing and acting, a problem as old as feedback control with dead time [62] and predictive tracking [4]. Classical catching and grasping systems estimate the target's trajectory with Kalman-type filters [5], servo on visual features [31], and plan an interception [2, 6, 26], later combined with grasp and reachability planning [1, 51]. These pipelines are precise but engineered for each object and task. Learned visuomotor policies [16, 95] and vision–language–action models (VLAs) [9, 11, 36, 53, 54, 103] generalize across objects and instructions, but they are trained and evaluated in quasi-static scenes, where the world waits while the model computes. Recent work brings learned policies to moving scenes. DOMINO provides a dynamic benchmark together with PUMA, which reads historical optical flow and is supervised with future object features [19]. DynamicManip synthesizes dynamic demonstrations from a single static one [40], DynamicWAM conditions a WAM on flow and kinematic descriptors of the past [48], and DynaWM refines a base VLA's action chunk for moving objects [14]. Other methods forecast salient visual tokens [65], a future image [92], a compact future code with the time to the next gripper event [49], or a 3D Gaussian velocity field that is optimized online [87], and feed the forecast to the policy at test time. Execution schemes reduce or hide the latency instead. DynamicVLA and StreamingVLA overlap inference with execution [58, 78], real-time chunking and bidirectional decoding keep consecutive chunks consistent [10, 45], asynchronous inference decouples prediction from control [60], VLASH rolls the robot state forward to the execution time [66], and Chronos Policy encodes observation and execution timestamps [70]. These schemes change when actions are computed and executed, whereas SE(3)-WAM changes what the policy represents, so the two are orthogonal. Closest to

ours, PUMA also uses future prediction only as a training signal. It forecasts mask-pooled image features of the target from 2D flow, whereas SE(3)-WAM forecasts the target's metric SE(3) trajectory and a sparse 3D flow of its surroundings.

World models and world–action models. World models learn how observations evolve, and they have been used for planning with visual foresight [20], for reinforcement learning in a learned latent space [23, 24], and for planning on pretrained visual features [3, 99]. In manipulation, predicted frames or subgoal images have been converted into actions [8, 17, 93], and video prediction has served as pretraining [76], as a predictive objective for inverse dynamics [68], and as a source of predictive features [28]. World–action models (WAMs) generate future video and actions jointly in one network [39, 72, 81, 100]. Recent studies compare how the two streams should exchange information [72, 83] and how long and how compressed their context should be [30, 46]. Because video is expensive and dominated by appearance, a growing body of work changes what is forecast: dynamic regions, depth, and semantic features [91], flow, geometry, and semantics alongside RGB [85], motion-weighted video [32], object masks [82] or object slots [47], optical flow [15] or motion vectors [41], compact future latents [37, 38], and targets in the policy's own condition or token space [64, 94]. A second question is whether the forecast is needed at test time. Fast-WAM finds that most of the benefit of video modeling remains when test-time imagination is skipped [86], and other methods drop their predictors [19, 64, 94] or auxiliary structure branches [85] after training. SE(3)-WAM follows this training-only use of the forecast but changes its target. Instead of pixels or latents at a uniform resolution, it forecasts a metric 4D state whose precision follows each entity's role: a full SE(3) trajectory for the target and a sparse 3D flow for the rest of the scene. Its action stream reads observation-derived summary tokens shaped by these forecasts, never the forecasts themselves, so the forecast heads are skipped at deployment.

Structured 3D and 4D representations. Metric 3D structure makes manipulation policies more precise and sample-efficient. Voxel, multi-view, and point-cloud policies act on scene geometry [21, 34, 59, 88], object-centric point clouds improve generalization [102], and 3D and 4D VLAs bring 3D and spatiotemporal inputs to large models [90, 96]. Motion has been exposed to policies through 2D point tracks [7, 74, 79, 97], dense pixel motion [52, 55, 56], and 3D flow [22, 84, 98], either as inputs or as intermediate predictions that are then turned into actions. Object-level representations include keypoints [50], object SE(3) trajectories [27, 63, 75], and separate encodings of the robot, the task object, and obstacles [18]. Rigid-body structure has long been used to learn scene dynamics, from SE(3) decompositions of point clouds [12, 13] and 3D dynamic scene memories [80] to scene flow estimation [44, 67], and PointWorld scales action-conditioned 3D point flow to in-the-wild robot data [29]. Point trackers, pose estimators, and feed-forward reconstruction models [33, 71, 73, 77, 89] now make such labels cheap to extract from video. These representations usually apply one granularity to the whole scene, either dense geometry and flow everywhere or poses for every object. SE(3)-WAM assigns granularity by role. The target, where contact depends on millimeters, is described by its SE(3) pose history and forecast, and the rest of the scene, which matters through the clearance it leaves, by a sparse 3D flow. For real robots, we pretrain on DROID with the 3D point tracks that PointWorld extracted from it [29].

3. SE(3)-WAM

SE(3)-WAM (Fig. 2) is a transformer with a world stream and an action stream that share attention. The world stream encodes the current observation and the recent motion and is trained to forecast a role-aware 4D scene. The action stream generates action chunks by conditional flow matching [42] and reads the world only through a few summary tokens. We first state the setting that motivates this design.

3.1. Acting on a Moving Target

Execution times. At query t, a policy receives an image, a depth map, and the robot state captured at time τ_t, together with histories of earlier states, and outputs a chunk $A_t = (\mathbf{a}_{t,0}, \dots, \mathbf{a}_{t,H-1})$ whose j-th command executes at

$$\tau^{\mathrm{a}}_{t,j} = \tau_t + \ell_t + j\Delta_a, \tag{1}$$

where ℓ_t is the latency from capture to the first execution (preprocessing, inference, and actuation) and Δ_a is the control period. A command must fit the scene at $\tau^{\mathrm{a}}_{t,j}$ rather than at τ_t (Fig. 3), and the lead $\Delta_{t,j} = \tau^{\mathrm{a}}_{t,j} - \tau_t$ grows with every millisecond of inference.

Target and context. Let $W_t(\mathbf{x}, \tau) \in \mathbb{R}^3$ be the displacement between τ_t and τ of the scene point at $\mathbf{x}$, in the robot base frame. The *target* $\mathcal{O}$ is the rigid body whose pose error enters a contact or containment tolerance ε, e.g., an object to grasp or a receptacle to place into. The *context* $\mathcal{C}$ contains the bodies that only need to be avoided, such as a belt or an occluder, with clearance m. On the target, the motion field is set by its pose $T(\tau) = (R(\tau), \mathbf{p}(\tau)) \in \mathrm{SE}(3)$:

$$W_t(\mathbf{x}, \tau) = T(\tau)\, T(\tau_t)^{-1}\mathbf{x} - \mathbf{x}, \qquad \mathbf{x} \in \mathcal{O}. \tag{2}$$

Where the error comes from. Let a policy act, explicitly or implicitly, on an estimate $\hat{T}$ of the target pose at τ^{a}. For a contact point at lever arm r and a rotation error θ, the contact

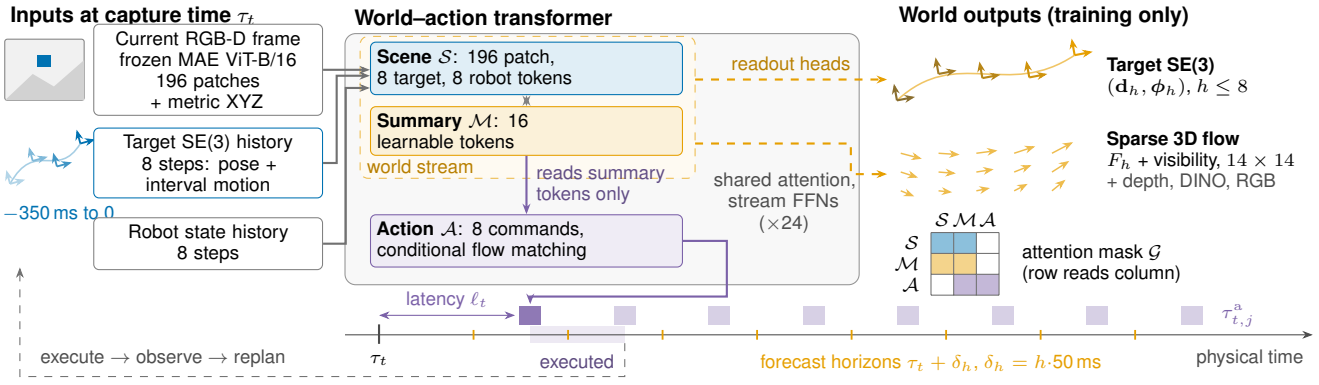


Figure 2. **SE(3)-WAM.** The world stream holds 212 scene tokens, the current image patches from a frozen encoder together with short histories of the target's metric SE(3) pose and of the robot state, and 16 summary tokens. During training, light heads read the world stream out into a role-aware 4D forecast at physical horizons $\tau_t + \delta_h$: the target's SE(3) trajectory and a sparse 3D flow with visibility on a 14×14 grid, plus appearance targets (dashed: training only). The action stream generates a chunk of 8 commands by flow matching. The attention mask $\mathcal{G}$ lets world tokens read only each other and action tokens read the summary tokens and each other, so the forecasts never reach the actions. The policy executes one command, observes, and replans.

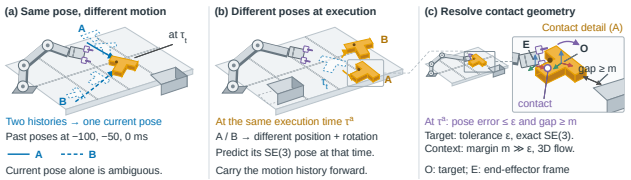


Figure 3. **Acting on a moving target** (schematic). (a) Two histories reach the same pose at capture time τ_t but imply different motions. (b) At the execution time τ^{a}, they lead to different target poses. (c) A contact requires the end-effector frame E to match the target O within ε, whereas the context matters only through the gap it leaves, $m \gg \varepsilon$.

error is at most $\|\hat{\mathbf{p}}(\tau^{\mathrm{a}}) - \mathbf{p}(\tau^{\mathrm{a}})\| + 2r\sin(\theta/2)$. Writing the motion over the lead Δ as constant velocity $\mathbf{v}$ plus a residual $\boldsymbol{\rho}$ (acceleration, rolling, impacts) gives

$$\|\hat{\mathbf{p}}(\tau^{\mathrm{a}}) - \mathbf{p}(\tau^{\mathrm{a}})\| \le e_{\mathrm{loc}} + \Delta\, e_v + \|\mathbf{v}\|\, e_\tau + e_{\mathrm{dyn}}, \quad (3)$$

the errors of the current position, of the velocity times the lead, of the timing times the speed, and of the residual motion (Appendix G). For the context, a collision requires a prediction error of at least m somewhere along the commanded path. Two conclusions follow. First, the precision that matters is concentrated on the target: a millimeter-level estimate of its pose and motion, including rotation through the lever arm r, whereas the context can be predicted far more coarsely. Second, latency enters the budget twice, through the lead Δ that multiplies every velocity error and through the motion that the residual accumulates, so information that costs inference time pays for itself only if it removes more error than the extra latency adds. These conclusions shape the three parts of SE(3)-WAM: what it remembers (Sec. 3.2), what it learns to forecast (Sec. 3.3), and how its actions use this forecast without paying for it at deployment (Sec. 3.4).

3.2. Inputs: The Current Frame and Motion Histories

SE(3)-WAM sees only one image. A frozen MAE ViT-B/16 [25] encodes the current RGB frame into 196 patch tokens, and the depth map gives each patch the metric 3D position of its center point in the base frame. The motion comes from two short histories over the last 8 control steps (350 ms at 20 Hz). The target history is the current target pose in the base frame together with the seven interval motions between consecutive samples: the translation, the relative rotation, the time interval, and the finite-difference velocity and acceleration, in physical units (Appendix A.1). It forms 8 target-state tokens. The robot history forms 8 end-effector state tokens. Together with the patch tokens, these are the 212 scene tokens. On the real robot, a causal tracker estimates the target pose in every frame: SAM 2 propagates a mask from a prompt on the target [57], and FoundationPose tracks the object from a model or a few reference views [73]. Each estimate is emitted once with its capture time, and later frames do not revise it (Appendix A.2). In simulation, we read the target pose from the simulator, which separates the policy from tracking errors, and we evaluate pose noise and tracked poses separately (Appendix F.2). The model receives no past images, so its input cost does not grow with the history.

3.3. Role-Aware 4D Forecast

The world stream is trained to forecast W_t at horizons $\tau_t + \delta_h$, with $\delta_h = h{\cdot}50$ ms and $h = 1, \ldots, 8$, which span the execution times of a whole chunk (400 ms). We parameterize

the target through Eq. (2). For each horizon,

$$\mathbf{d}_h = \mathbf{p}(\tau_t+\delta_h) - \mathbf{p}(\tau_t), \quad \boldsymbol{\phi}_h = \mathrm{Log}\big(R(\tau_t+\delta_h)R(\tau_t)^\top\big)^\vee, \tag{4}$$

and a pose head regresses $(\mathbf{d}_h, \boldsymbol{\phi}_h)$ with a squared translation error plus λ_R times the squared geodesic rotation error ($\mathcal{L}_P$). The rest of the scene is a sparse 3D flow: for the center point of each of the 196 patches on the 14×14 grid, about [X] mm apart in the workspace, a flow head regresses its base-frame displacements $F_h \in \mathbb{R}^3$ and its visibility at every horizon ($\mathcal{L}_F$). The grid is deliberately coarse, since the context is tolerated up to its clearance, and the target's precision comes from its pose rather than from its points. As further training targets, appearance heads regress the future depth, the features of a frozen DINOv3 teacher [61], and the RGB of moving surfaces, and a reconstruction head restores the target region of the current image ($\mathcal{L}_{\text{aux}}$). All heads are light readouts of the final hidden states of the world stream, all labels are cumulative from the current state, and the losses use valid labels only.

3.4. Actions and the Shared Backbone

The world stream holds the 212 scene tokens and 16 learnable summary tokens. The action stream holds 8 tokens, one per command, each embedding a noisy command and the flow time. A chunk spans 8 control periods of $\Delta_a = 50$ ms, and each command is an end-effector translation and rotation plus a gripper command. The attention mask $\mathcal{G}$ lets world tokens attend only to world tokens and lets action tokens attend to the summary tokens and to each other. The actions therefore read the world only through the summary tokens, whose content the forecast losses shape during training, while the forecasts themselves never enter the action stream. Both streams share the attention projections of each layer and have separate feed-forward networks, so they exchange information in attention but keep stream-specific capacity. For a normalized chunk $\mathbf{a}_1$, noise $\mathbf{a}_0 \sim \mathcal{N}(0, I)$, and $u \sim \mathcal{U}[0, 1]$, the action loss is

$$\mathcal{L}_A = \mathbb{E}\left\|v_\psi(\mathbf{a}_u, u \mid \mathcal{S}_t, \mathcal{G}) - (\mathbf{a}_1 - \mathbf{a}_0)\right\|^2, \tag{5}$$

with $\mathbf{a}_u = (1-u)\mathbf{a}_0 + u\mathbf{a}_1$ and the scene tokens $\mathcal{S}_t$ of query t. The total loss is $\mathcal{L} = \mathcal{L}_A + \lambda_F\mathcal{L}_F + \lambda_P\mathcal{L}_P + \lambda_{\text{aux}}\mathcal{L}_{\text{aux}}$. Sec. 4.4 compares this coupling with alternatives, which we label as follows: R0 disables the forecast losses; R2 adds world queries whose hidden states the actions read at every layer; R3 lets these queries read the actions, and R4 opens both directions; and R5 encodes the decoded forecast as geometric tokens for the actions (Appendix A.5).

Deployment. At test time, the forecast heads are skipped. Because world tokens never attend to actions, the world stream runs once per query, and the action stream integrates 5 Euler steps on its cached keys and values. The robot executes the first command, the policy observes again, and it replans, so every executed command is generated from an observation taken only ℓ_t earlier. The p95 time-to-action is [XX] ms on an RTX 5090.

Implementation. The transformer has 24 layers of width 1024 with 16 heads and feed-forward width 4096, with 504.0M backbone parameters and 523.8M trainable parameters in total. The MAE ViT-B/16 encoder (85.8M) and the DINOv3 teacher (303M) stay frozen. On Dynamic-14, we train on all windows of 800 demonstrations per task for 20 epochs (51,740 updates at a global batch size of 512) with AdamW and a cosine schedule, with $\lambda_F = \lambda_P = 1$, $\lambda_R = \lambda_{\text{aux}} = 0.1$, which takes [X] hours per training seed on [N] RTX PRO 6000 GPUs (Appendix A.6).

3.5. Pretraining for Real Robots

Real robot data do not provide object poses, so the real-robot base learns the parts of the forecast that DROID [35] supports. It has the same architecture with 48 layers (1.02B trainable parameters) and the same inputs except the target history. Its targets are the actions, the sparse 3D flow, and the future DINOv3 features. The flow labels come from the 3D point tracks and the optimized camera calibrations that PointWorld released for DROID [29], resampled to one track per image patch and to DROID's native timing. We train on 598,889 windows from 42,087 episodes for 21,060 updates on 64 GPUs (Appendix B). Each real-robot policy initializes from this base, adds the target-state tokens and the pose head, and is fine-tuned on its task with tracked poses. The simulation model is trained from the generic initialization, a frozen MAE encoder with a randomly initialized transformer.

4. Experiments

We compare SE(3)-WAM with reactive, dynamic, and world–action policies in simulation and on a real robot (Secs. 4.2 and 4.3) and then ablate its history, its forecast, and the way its actions use the forecast (Sec. 4.4).

4.1. Setup

Dynamic-14. We built fourteen tasks in MuJoCo/robosuite [69, 101] in five families: grasping a moving target (harmonic, conveyor, translating, rotating, and occluded motion), placing into a moving receptacle, passing a wand along a rotating wire, capturing rolling or sliding objects, and grasping after an impact (Appendix C, Fig. S3). Sweeps vary the speed (0.5–1.5×), the added latency (0–200 ms), and the passage clearance ([values] mm). Experts with privileged simulator state provide 800 demonstrations per task. We select configurations on development scenes and report every Dynamic-14 number on 1,400 held-out scenes (100 per task, new seeds). Our variants average three training seeds, and

each baseline is trained once. We give paired 95% intervals from a bootstrap over held-out scenes.

Protocol. Observation, scene motion, and execution share one physical clock with a control period of 50 ms. All methods run under the same executor, and physics advances while a method computes, by its measured latency on one RTX 5090. A policy is queried again once a command has executed and its previous inference has finished, and until a new chunk arrives, the executor continues with the previous one. The +150 ms setting adds 150 ms of delay for every method, and the paused setting stops physics during inference (Appendix C.3). We report the task-balanced success rate (SR), the contact-time error e_t, the position error at first contact e_p, and the p95 time-to-action.

Baselines. We compare with reactive policies (ACT [95], Diffusion Policy [16], and $\pi_{0.5}$ [54]), dynamic VLAs (DynamicVLA [78] and PUMA [19]), image- or latent-space WAMs (DreamVLA [91] and OpenWAM-α [72]), and a dynamic WAM (DynamicWAM [48]). Each baseline uses its official code, inputs, and recipe, is trained on the same demonstrations, and receives a rendered wrist camera if its recipe uses one (Appendix D). SE(3)-WAM uses only the agent view, but it also uses depth and the target pose. To separate these inputs from the model, $\pi_{0.5}$ and DynamicWAM also receive our SE(3) history tokens (+SE(3)), and two baselines use the pose history alone: *CV input* is SE(3)-WAM without forecast losses that receives the constant-velocity extrapolation of the target pose as input tokens, and a Kalman interceptor plans to the target pose predicted by a constant-acceleration Kalman filter [5].

4.2. Main Results

SE(3)-WAM reaches [XX.X]% SR on held-out Dynamic-14 scenes (Table 1), [X.X] points above the strongest baseline, [name of the strongest baseline], and [X.X] points above the strongest baseline that receives the same SE(3) history, [name]. The margin is largest on [rotating targets and on fast conveyor and translating targets], where the target moves farthest during the lead. Its p95 time-to-action of [XX] ms is [X]$\times$ lower than that of the image-space WAMs, whose latency adds to the lead they must bridge. With 150 ms of added delay, SE(3)-WAM keeps [XX.X]%, whereas reactive policies drop to [XX.X]% and image-space WAMs to [XX.X]%. With physics paused during inference, the difference to [name of the strongest paused baseline] is [$\pm$X.X] points, which separates the contribution of speed from that of the representation. The CV-input variant and the Kalman interceptor trail SE(3)-WAM by [X.X] and [X.X] points, mostly on non-constant motion, so the learned model does more than extrapolate the pose. Results on DOMINO [19] and on static LIBERO tasks [43] are in Appendix D.2.

Figure placeholder: Fig. 4 — real-robot trials

Rows: a success of SE(3)-WAM on the accelerating task; the strongest baseline on the same trial key; a failure of SE(3)-WAM. **Columns:** the query frame, the first execution time τ^{a}, and contact. **Overlays:** blue tracked past poses; purple commanded gripper path. Real photos, unretouched; overlays from logged data of each trial.

Figure 4. **Real-robot trials.** SE(3)-WAM closes where the target will be when the command executes, whereas the baseline closes behind it. Bottom: a failure after the tracker lost the target under the gripper.

4.3. Real-Robot Experiments

We evaluate five tasks on a seven-DoF Franka arm with a parallel-jaw gripper and a calibrated RGB-D camera (Fig. S4). The target moves along a line in two tasks ([T1, T2]), on a circle in two tasks ([T3, T4]), and with acceleration in one task ([T5]) (Table 2, Appendix D.3). Each method is fine-tuned per task on the same [N] teleoperated demonstrations and runs [20] trials per task with shared trial keys. SE(3)-WAM tracks the target with FoundationPose [73] and runs on one RTX 5090 with a p95 time-to-action of [XX] ms. It succeeds in [XX.X]% of all trials, against [XX.X]% for [the strongest baseline] and [XX.X]% without the DROID base (Table 2). Its advantage is largest on [accelerating motion], where extrapolating the last observed velocity fails, and its remaining failures are mostly [late closures after tracking losses under the gripper] (Fig. 4).

Pretraining. The DROID base raises the real-robot SR by [X.X] points over the generic initialization (Table 2), and a base pretrained with actions only recovers [XX]% of this gain at the same budget, so most of the benefit comes from the 3D flow and DINO targets (Appendix B).

4.4. Ablations: What Does a Dynamic Policy Need?

Each variant changes one component of the final model and keeps the data, the training budget, and the other components fixed (Table 3 and Fig. 5). The complete comparisons are in Appendix F.

What to remember. We keep the current frame, the current target pose, and the robot history fixed. Adding eight frames of RGB, optical flow over the same window with DynamicWAM's motion statistics [48], or video-VAE latents changes SR by [X.X] to [X.X] points and raises the time-to-action by up to [X]$\times$, whereas the SE(3) history adds [X.X] points at almost no cost (Table 3). The gain appears once the lead exceeds the critical lead $\varepsilon/\|\mathbf{v}\|$ of a family, grows with

Table 1. **Main results.** Held-out SR (%) on Dynamic-14 (1,400 new scenes, task-balanced, real time, three training seeds for our variants and one for baselines) by family, with 150 ms of added delay, and with physics paused during inference, together with the contact-time error e_t (ms), the position error at first contact e_p (mm), and the p95 time-to-action on one RTX 5090. Inputs: R, RGB and robot state. D, depth. P, target SE(3) history. Best in bold. Intervals in Appendix E.

		Dynamic-14 SR, real time unless noted										
Method	Inputs	All	Grasp	Place	Pass	Capture	Impact	+150 ms	Paused	e_t	e_p	p95 (ms)
ACT [95]	R	—	—	—	—	—	—	—	—	—	—	—
Diffusion Policy [16]	R	—	—	—	—	—	—	—	—	—	—	—
$\pi_{0.5}$ [54]	R	—	—	—	—	—	—	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	R+P	—	—	—	—	—	—	—	—	—	—	—
DynamicVLA [78]	R	—	—	—	—	—	—	—	—	—	—	—
PUMA [19]	R	—	—	—	—	—	—	—	—	—	—	—
DreamVLA [91]	R	—	—	—	—	—	—	—	—	—	—	—
OpenWAM-α [72]	R	—	—	—	—	—	—	—	—	—	—	—
DynamicWAM [48]	R	—	—	—	—	—	—	—	—	—	—	—
DynamicWAM + SE(3)	R+P	—	—	—	—	—	—	—	—	—	—	—
SE(3)-WAM, CV input	RD+P	—	—	—	—	—	—	—	—	—	—	—
Kalman interceptor	D+P	—	—	—	—	—	—	—	—	—	—	—
SE(3)-WAM	RD+P	—	—	—	—	—	—	—	—	—	—	—

Figure placeholder: Fig. 5 — ablations (four panels)

(a) History. x: added latency (0–200 ms); y: ΔSR vs. current frame only. Lines: RGB window, optical flow, video-VAE latent (gray), SE(3) history (blue), SE(3) history with pose noise (blue dotted); stationary targets as a faint line; ticks at the families' critical lead $\varepsilon/\|\mathbf{v}\|$. **(b) Target forecast.** Bars by motion type (constant velocity, harmonic, rolling/sliding, post-impact): forecast error at τ^{a} of CV, CA, and the learned SE(3) forecast (orange), with SR of the corresponding policies as dots. **(c) Context.** Passage tasks only. x: clearance (mm); y: SR; lines: no context flow, 14^2, 28^2, 56^2 grids, future latent; markers: each grid's measured path error. **(d) Coupling.** x: p95 time-to-action (ms); y: SR at 0 and 150 ms added latency; points: R0, SE(3)-WAM, R2, R4, R5. 95% paired bootstrap bands, held-out scenes, three seeds.

Figure 5. **Ablations.** (a) The SE(3) history helps once the lead exceeds the critical lead, and more with latency, while visual histories stay flat. (b) The learned SE(3) forecast beats extrapolation on non-constant motion. (c) A coarse context flow suffices until the clearance reaches its path error. (d) Reading or decoding the forecast at test time adds latency but not success.

Table 2. **Real-robot results.** Success rate (%) on five tasks, grouped by the target's motion, [20] trials per task and method, and over all trials. Every method is fine-tuned per task on the same demonstrations. Binomial 95% intervals are given in Appendix D.3.

	Linear		Circular		Accel.	
Method	[T1]	[T2]	[T3]	[T4]	[T5]	All
$\pi_{0.5}$ [54]	—	—	—	—	—	—
DynamicWAM [48]	—	—	—	—	—	—
SE(3)-WAM, no DROID base	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—

latency, and vanishes for stationary targets (Fig. 5a). Frozen readouts show that visual histories do contain the motion (Appendix F.2), but at our data scale the policy does not turn it into metric velocity. With pose noise of a real tracker, the SE(3) history keeps [XX]% of its gain. The same history given to $\pi_{0.5}$ and DynamicWAM raises their SR by [X.X] and [X.X] points (Table 1), so the benefit is not specific to our model.

What to predict. Disabling all forecast losses (R0) costs [X.X] points, and the gap grows with the lead. Forecasting only the target's SE(3) trajectory costs [X.X] points, mostly on passage and placement tasks, where the context constrains the motion, and forecasting only the context flow costs [X.X], mostly on rotating targets, where the lever arm amplifies rotation errors. Replacing the target's pose by the flow of its points, at the same uniform granularity as the context, costs [X.X] points. At the execution times, the learned SE(3) forecast is as accurate as constant-velocity extrapolation on conveyor motion but [X] times more accurate on rolling,

Table 3. **Ablations** on held-out Dynamic-14 scenes (three training seeds). Each block changes one component of SE(3)-WAM (⋆). SR, SR_{150}: success (%) without and with 150 ms of added delay. e_t: contact-time error (ms). e_p: position error at first contact (mm). p95: time-to-action (ms).

Variant	SR	SR_{150}	e_t	e_p	p95
SE(3)-WAM ⋆	—	—	—	—	—
History (current frame and pose kept)					
none	—	—	—	—	—
RGB window (8 frames)	—	—	—	—	—
optical flow, motion statistics	—	—	—	—	—
video-VAE latent	—	—	—	—	—
SE(3) with tracker noise	—	—	—	—	—
Forecast					
none (forecast losses off, R0)	—	—	—	—	—
target SE(3) only	—	—	—	—	—
context flow only	—	—	—	—	—
uniform: point flow on target	—	—	—	—	—
image space: future RGB/latent	—	—	—	—	—
no appearance targets	—	—	—	—	—
Use of the forecast at test time					
action reads world queries (R2)	—	—	—	—	—
world reads actions (R3)	—	—	—	—	—
bidirectional (R4)	—	—	—	—	—
action from decoded forecast (R5)	—	—	—	—	—

Figure placeholder: Fig. 6 — forecasts at the execution time

Rows: a rolling capture (median forecast error) and a post-impact grasp that fails. **Columns:** the query frame at τ_t, the frame at τ^a, and a 3D inset in the base frame. **Overlays:** blue past poses, orange forecast triads and coarse flow arrows from the training-time heads, gray ground truth at the same times, purple commanded path. Full set in Fig. S5.

Figure 6. **Forecasts at the execution time** on held-out Dynamic-14 episodes, read from the training-time heads. Top: the SE(3) forecast follows a rolling target that extrapolation overshoots. Bottom: before an impact, the regressed forecast averages the possible outcomes, and the grasp misses.

sliding, and post-impact motion (Fig. 5b). For the context, the 14×14 flow, with a path error of [X] mm, matches 28^2 and 56^2 grids until the passage clearance falls to [X] mm (Fig. 5c). Image-space futures, i.e., future RGB or video latents in place of the 4D forecast, reach [XX.X]% at [X]× the training cost. As auxiliary targets next to the 4D forecast, however, depth, DINO features, and moving-region RGB add [X.X] points, and like the 4D heads they are skipped at deployment.

How to use the forecast. Letting the actions read the hidden states of world queries (R2), letting the world also read the actions (R3, R4), or generating the actions from the decoded forecast (R5) changes SR by [±X.X] to [±X.X] points, within the seed variation, at [X.X]–[X.X]× the time-to-action of SE(3)-WAM (Fig. 5d). Under 150 ms of added delay, the differences stay within [±X.X] points. The forecast thus helps as a training signal that shapes what the actions read, as Fast-WAM found for static scenes [86], and generating it at test time adds latency without adding success. Separate parameters for the world and action streams instead of shared attention change SR by [±X.X] points at matched size (Appendix F.1).

5. Conclusion

We presented SE(3)-WAM, a world–action model for manipulating moving objects that spends precision where contact depends on it and time only where it pays off. It remembers the target's metric SE(3) motion instead of past frames, learns to forecast the target's SE(3) trajectory precisely and the rest of the scene as a sparse 3D flow, and lets its actions read a representation shaped by these forecasts without generating them at test time. On Dynamic-14, where the scene keeps moving while a policy computes, and on five real-robot tasks, it improves on reactive, dynamic, and world–action policies. Our ablations attribute the gains to the metric motion history and to role-aware forecasts used as a training signal, rather than to richer appearance models or to reading the forecast online. For world–action models in dynamic settings, they favor forecasting the few quantities that decide contact and keeping the deployed policy as fast as a reactive one.

Limitations. SE(3)-WAM assumes a rigid target that moves on its own until contact, so deformable or articulated targets and post-contact dynamics are outside its scope. It depends on a pose estimate of the target, and tracker failures propagate to the policy; in simulation we read the pose from the simulator. The regressed target forecast cannot represent multimodal futures, e.g., before an impact. Our real-robot base learns from DROID without target poses, so its pose-related parts are learned only during fine-tuning. Finally, our tasks have a single target and slow context motion.

LLM Usage Statement

[To be completed before submission: actual tools used, affected parts of the research and manuscript, and extent of use.]

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